Making HEPA's bound empirical: an a priori test for whether HEPA will work on a new dataset

Aviral Garg · Forgis research proposal · 10 October 2026 · on HEPA (Petersen et al., arXiv:2605.11130v4)

The gap I want to close, stated by the paper itself

HEPA's Proposition 1 gives $I(H_t; E_{t+\Delta t}) \geq I(H^*; E_{t+\Delta t}) - C_\eta L^2 \varepsilon$, with vacuity threshold (their eq. 17) $\varepsilon_{\text{vac}} = I(H^*; E_{t+\Delta t})/(C_\eta L^2)$.

- Within a dataset the bound is validated: varying ε via pretraining snapshots gives Spearman $\rho = -0.87$ (C-MAPSS-1), -0.67 (C-MAPSS-3), -0.64 (MBA), -0.49 (SMAP).
- Across datasets it falls apart. Pooling converged ε over the 14 benchmarks gives Pearson $r = -0.05$ ($p = 0.90$). The paper attributes this to L, C_η and target scale differing by dataset.
- The constants are never estimated. In the authors' words: "The constants L and C_η are not estimated directly; fig. 3 validates only the monotonic relationship, not the full quantitative bound," and in Future Work: "deriving fully empirical versions of the information-retention bound that estimate L and C_η directly from data remains an interesting open problem."

As it stands the bound only explains HEPA's failures after the fact. I want to make it predictive. Define the dimensionless slack

$$\hat{\rho}(\mathcal{D}) \;=\; \frac{\hat{\varepsilon}}{\hat{\varepsilon}_{\text{vac}}} \;=\; \frac{\hat{\varepsilon} \cdot \hat{C}_\eta \hat{L}^2}{\hat{I}(H^*; E)}, \qquad \text{bound non-vacuous} \iff \hat{\rho} < 1.$$

Hypotheses

- H1 (normalisation restores the cross-dataset signal). Across the 14 benchmarks, $\rho_s(\hat{\rho},\ \text{h-AUROC}) \leq -0.6$, where raw ε gives $r = -0.05$. Falsified if $|\rho_s| < 0.3$, which would mean the incommensurability comes from somewhere else and my normalisation is wrong.
- H2 (it works as a screen). $\hat{\rho} \geq 1$ identifies the no-guarantee regime using no downstream result: CHB-MIT (h-AUROC 0.497, chance) and GECCO (within-dataset $\rho = +0.14$, $p = 0.67$) satisfy $\hat{\rho} \geq 1$; C-MAPSS-1 (0.806) satisfies $\hat{\rho} \ll 1$. Falsified if either control is misclassified.
- H3 (the L1/L2 mismatch is measurable). Assumption A2 is stated in L2, $\mathbb{E}\|\hat{H} - H^*\|_2^2 \leq \varepsilon$, while HEPA's eq. (2) minimises L1. So the bound is about a quantity the training doesn't optimise. Prediction: $\hat{\rho}$ from the L2 residual rank-correlates with h-AUROC better than $\hat{\rho}$ from the L1 residual, by $\Delta\rho_s \geq 0.15$. Falsified if the two sit within noise of each other, which would still be useful: it would license the L1 loss.

Estimators, all cheap and low-dimensional where it matters

Fit HEPA's own linear hazard head $\hat{\eta}(h) = \sigma(w^\top h + b)$ (513 params) on the frozen target representations h^*. Then, on held-out data:

- $\hat{L} = Q_{0.99}\big\{\, |\hat{\eta}(h_i) - \hat{\eta}(h_j)| \,/\, \|h_i - h_j\|_2 \;:\; (i,j) \in k\text{NN} \,\big\}$, using a high quantile rather than a supremum so the estimate stays stable at finite sample size.
- $\hat{C}_\eta = \big(2\,\underline{\eta}\,(1-\overline{\eta})\big)^{-1}$ with $\underline{\eta}, \overline{\eta}$ the 1st/99th percentiles of $\hat{\eta}(h^*)$.
- $\hat{I}(H^*; E) = H(\pi_e) - \mathbb{E}\big[H(\hat{\eta}(H^*))\big]$, plug-in. E is binary, so this is a 1-D estimate. No high-dimensional MI estimator is needed, which is usually where this kind of proposal dies.
- $\hat{\varepsilon} = \mathbb{E}\|\hat{h} - h^*\|_2^2$ on a held-out split, per H3 also in L1.

Why it is worth doing, and what it costs

- Scientific: it turns HEPA's central theorem into something you can compute and falsify, and it answers the open problem the paper names itself.
- Industrial: $\hat{\rho}$ needs unlabelled pretraining and a small labelled probe, so Forgis can tell a customer before committing whether HEPA will work on their plant. That's the deployment question.
- Cost: released checkpoints and HEPA's own recipe. The paper reports pretraining at under a minute per dataset on one A10G, and the new compute is $O(N)$ estimators plus one kNN pass. This fits well inside the ICML timeline.

Risks I can already name

- Plug-in MI is biased. Mitigation: report rank correlation (invariant to monotone bias) with bootstrap CIs, never absolute slack.
- $\hat{L}$ is a tail statistic and will be noisy. Mitigation: report the full sensitivity curve across the quantile rather than a single value.
- $n = 14$ datasets is low power. Mitigation: pre-register the predicted direction and the exact test before computing any h-AUROC correlation, and show the per-dataset scatter alongside the coefficient.

Secondary direction (from my analysis of `embeddings.npz`)

In the attached report I found that with a frozen target, no SIGReg weight $\alpha \in [0.01, 0.9]$ buys both prediction quality and representation rank (at HEPA's own α=0.1: cosine 0.8715±0.0003 → 0.066±0.007, 5 seeds). HEPA escapes this only because both encoders train jointly, which lets the target migrate toward the isotropic Gaussian SIGReg demands. So SIGReg is doing two jobs at once: it prevents collapse, and it sets the target's distribution.

- H4. Split the two jobs: SIGReg on the target branch, and an explicit rank floor on the predictor output. This removes the sensitivity to α and helps where channel-fusion tokenisation currently costs HEPA accuracy (the paper names BATADAL and MBA as having "sensor-localised events where channel-fusion tokenisation dilutes the relevant subset"). Measurable mediator: predictor-output effective rank.

Open question for our call: the paper reports α=0.1 (table 7), while `Forgis-Labs/HEPA` PR #3 states the table-1 runs used 0.04. I've worked from 0.1 throughout, since that's what the paper says. Which should a reimplementation use?